Temporal Order Changes a Bounded-Memory Information–Recovery Frontier

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Abstract

Can the temporal order of the same linear predicate change the operational power of a fixed quantum memory? A verifier streams four halves of independent EPR pairs through a device, sequestering each output before the next input. Only after commitment does a late choice request either the complete two-bit syndrome (AUDIT) or coherent recovery of all four pairs (RETURN). The device has one persistent qubit and unlimited genuinely classical transcript. Two check matrices contain the same columns: the grouped order (e_1, e_1, e_2, e_2) and the interleaved order (e_1, e_2, e_1, e_2). The grouped order attains the full static qubit frontier and permits return fidelity 1/2 at perfect audit. The interleaved order is instead bounded by, and attains, 1/4. More generally, m consecutive full-rank syndrome blocks impose a temporal product law at perfect audit. The main result makes that law robust: if D is the input dimension and $\epsilon = 1 - P_{\mathrm{A}}$, the spectral mass below the perfect-audit rank is at most $mD\epsilon$. Causal dimension-to-list conversion, Schur–Horn majorisation, and flagged polar recovery then give an explicit approximate-audit bound. For the interleaved instance this certifies a strict order-dependent support gap for every $3/7 < \lambda < 1$, including balanced weight. A local Pauli-completion theorem also reduces the arbitrary adaptive optimum exactly to one normalised bond-two Choi MPS. At $\lambda = 3/5$, a rational four-effect construction and a finite computer-assisted certificate enclose the exact streamed support between 0.7658988152 and 0.76670. Replaying the conic certificates calls no optimiser; the full support curve and exact equality at this direction remain open.

Keywords: bounded coherent memory; temporal order; quantum combs; entanglement recovery; syndrome decoding; Ky Fan norms.

1 Introduction

Memory dimension is a natural resource in a multitime quantum process, but dimension alone does not say how difficult a temporal task is. A memory may need to preserve several partial functions simultaneously, and the number of live functions can change when otherwise identical inputs are reordered. This work isolates that phenomenon in a late-choice information–recovery game.

The verifier sends fresh entangled carriers one at a time. A device must commit each output before seeing the next input and does not learn the terminal task in advance. AUDIT rewards a classical syndrome of the stream, whereas RETURN rewards preservation of the unknown quantum inputs. A coherent memory can accumulate a syndrome reversibly; a classical record that helps AUDIT also damages RETURN. The question is whether a mere permutation of the syndrome columns changes the best tradeoff when the coherent memory is fixed.

It does. The pair in Fig. 1 uses the same rank-two binary code and the same one-qubit memory. Grouping equal columns realizes the complete static qubit frontier. Interleaving them creates two consecutive full-rank temporal blocks and lowers perfect-AUDIT return fidelity from 1/2 to 1/4. The effect is not limited to an endpoint: a robust rank-tail theorem proves a strict support-function separation throughout $3/7 < \lambda < 1$.

The remaining interior optimisation can be stated without an unbounded adaptive tree. Every normalised bond-two Choi leaf admits a local four-outcome Pauli completion whose transcripts are covariance-related copies of that leaf. Consequently the exact support is a compact Choi-MPS maximum. This reduction turns local completeness from an obstruction into a theorem; it does not make the resulting nonconvex maximum analytic automatically. A finite arity split nevertheless closes one nontrivial support direction to width 0.0008011848, while keeping the numerical and formal trust boundaries explicit.

Main consequence. For the interleaved stream, every admissible strategy obeys $\beta_{H_{\mathrm{I}},2}(\lambda) \le U_{\mathrm{I}}(\lambda)$, and $U_{\mathrm{I}}(\lambda) < B_{4,2}(\lambda)$ whenever $3/7 < \lambda < 1$. Thus temporal order changes the achievable information–recovery region in the interior, not only at perfect AUDIT. At $\lambda = 3/5$, exact replay further gives $0.7658988152 \le \beta^{\mathrm{stream}}_{H_{\mathrm{I}},2}(3/5) \le 0.76670$.

The broad ingredients are established. Quantum combs and process tensors describe multitime strategies and their internal memories [1, 2, 3]. Bounded coherent memory has been studied through constrained-

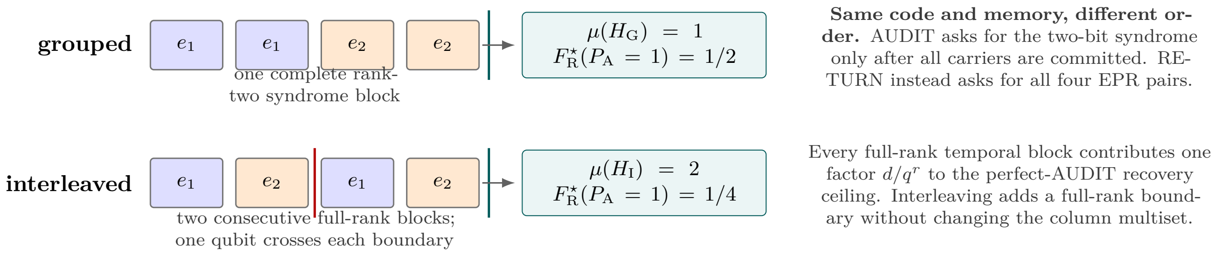

Figure 1: The same four columns in two temporal orders. The grouped stream has one complete rank-two block. The interleaved stream has two; a one-qubit bond must cross both full-rank boundaries. AUDIT sees only the classical transcript and terminal qubit, whereas RETURN regains all committed carriers.

separability hierarchies and autonomous process discrimination [4, 5]. Postmeasurement-information games allow unlimited classical data beside a bounded quantum register [6, 7], while information–disturbance theory relates measurement records to later recovery [8, 9, 10]. Classical trellis complexity is famously order dependent [11, 12, 13]. The claim here is narrower: an exact same-code temporal-order separation in a common-prefix syndrome-AUDIT/all-carrier-RETURN game, together with a causal-list rank-tail bound. A focused search through 27 August 2026 located no direct antecedent of that conjunction; this is a dated collision assessment through 27 August 2026, not proof of priority.

2 Late-choice streaming model

At slot $i \in \{1, \dots, n\}$, the verifier prepares

$$|\Phi^+\rangle_{R_i A_i} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \tag{1}$$

sends A_i to the device, receives B_i, and immediately sequesters B_i. The device applies an arbitrary adaptive instrument to A_i, fresh local ancillas, and a persistent memory M_{i-1}. Only M_i, with declared dimension at most d_i, may remain coherent across the next cut. An unrestricted finite classical transcript is free, but it must be genuinely dephased: retaining its coherent purification would be an additional quantum memory.

After slot n, a hidden choice selects one task. In AUDIT, the references are measured in the computational basis, producing a uniform word $X \in \mathbb{F}_q^n$. The device receives neither references nor sequestered carriers and must output the full syndrome $HX \in \mathbb{F}_q^r$ using only its transcript and terminal memory. Its unconditional success is $P_{\rm A}$. In RETURN, a transcript-conditioned decoder acts jointly on all B_i and the terminal memory, without reference access. The verifier tests all EPR pairs and the charged memory reset. Its unconditional acceptance probability is $F_{\rm R}$.

For $0 \le \lambda \le 1$, define the support function

$$\beta_{H,d}^{\rm stream}(\lambda) = \sup_{\mathcal{S}} \big[\lambda P_{\rm A}(\mathcal{S}) + (1-\lambda) F_{\rm R}(\mathcal{S})\big], \tag{2}$$

where the supremum includes arbitrary finite-outcome adaptive non-QND strategies satisfying the interface. No postselection is allowed. Immediate sequestration matters: past carriers cannot silently enlarge a later memory.

For the canonical binary pair,

$$H_{\rm G} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \qquad H_{\rm I} = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}. \tag{3}$$

They differ only by a coordinate permutation. Both have four equiprobable syndromes, and the persistent memory is one qubit.

3 Static ceiling and grouped attainment

Ignoring streaming order gives a useful terminal-dimension relaxation. For a four-label ensemble and a terminal qubit, the exact support is

$$B_{4,2}(\lambda) = \frac{1 + \sqrt{\lambda^2 + (1-\lambda)^2}}{2}. \tag{4}$$

The bound follows by refining the classical flag, applying the dimension-two minimum-error bound to AUDIT, and the flagged recovery inequality to RETURN. For a normalized four-component likelihood row, the extremal profile has two equal large entries and two equal small entries. The remaining optimization is the Euclidean norm in (4).

Proposition 3.1 (Grouped frontier)**.** *For $H_{\rm G}$ and a persistent qubit,*

$$\beta_{H_{\rm G},2}^{\rm stream}(\lambda) = B_{4,2}(\lambda) \quad (0 \le \lambda \le 1). \tag{5}$$

An exposed parametrization is

$$P_{\rm A} = \frac{1+t}{2}, \qquad F_{\rm R} = \frac{1+\sqrt{1-t^2}}{2}, \qquad 0 \le t \le 1. \tag{6}$$

The device coherently accumulates the first parity, weakly records it with strength t, uncomputes the carrier-dependent part, and hands the same qubit to the second parity. The classical flag and terminal qubit identify the full syndrome in AUDIT; the carrier outputs undo the coherent updates in RETURN. At perfect AUDIT, (6) gives $F_{\rm R} = 1/2$.

4 Temporal fragmentation at perfect audit

The endpoint extends well beyond the four-slot example. Let $H \in \mathbb{F}_q^{r\times n}$ be split into m consecutive blocks,

$$H = [H^{(1)}|\cdots|H^{(m)}], \qquad \operatorname{rank} H^{(j)} = r. \tag{7}$$

Put $N = q^r$. Let at most $d_j \le N$ coherent dimensions cross the boundary after block j, including the terminal boundary. The transcript remains free and classical.

Theorem 4.1 (Perfect-AUDIT temporal product law)**.** *Under* (7),

$$P_{\mathrm{A}} = 1 \implies F_{\mathrm{R}} \le \prod_{j=1}^{m} \frac{d_j}{N}. \tag{8}$$

For uniform $d = q^s$, the bound is attained on $H = [I_r|\cdots|I_r]$ by retaining s syndrome coordinates coherently and recording the other $r-s$ coordinates in each block.

Each full-rank block introduces an independently uniform syndrome contribution. Perfect terminal decoding constrains every refined Kraus leaf to at most d_j cumulative labels at boundary j. Intersecting those constraints leaves a support fraction $\prod_j (d_j/N)$. Optimal flagged recovery converts that rank fraction into (8).

For H_{G}, one can pack only one consecutive full-rank block, so the endpoint is $1/2$. For $H_{\mathrm{I}} = [I_2|I_2]$, there are two and the endpoint is $1/4$. Both are attained. Repeating each basis column m times gives a larger same-columns separation: batching equal columns yields d/q^r, while cycling through the basis m times yields $(d/q^r)^m$. The ratio grows as $(q^r/d)^{m-1}$. Consecutive trellis sectionalization itself is classical prior art [14]; the new statement is the late-choice recovery law attached to it.

5 Causal lists and a linear spectral tail

Perfect decoding forces exact low rank. Approximate decoding requires a quantitative substitute. Let $D = q^n$ and

$$\alpha = \prod_{j=1}^{m} \frac{d_j}{N}, \qquad k = \alpha D = q^{n-mr} \prod_j d_j. \tag{9}$$

Refine every instrument outcome to a single Kraus label and reveal the label to both late decoders. This only enlarges the strategy class. A complete transcript c has a sequential Kraus operator K_c and effect $G_c = K_c^\dagger K_c$. Let

$$t_c = \sum_{\ell > k} \lambda_\ell(G_c), \tag{10}$$

where eigenvalues are nonincreasing.

Theorem 5.1 (Linear rank-tail robustness)**.** *For every admissible strategy under* (7)*, with $\epsilon = 1 - P_{\mathrm{A}}$,*

$$\boxed{\sum_c t_c \le mD\epsilon.} \tag{11}$$

Consequently, for $\theta = \min\{m\epsilon, 1-\alpha\}$,

$$\boxed{F_{\mathrm{R}} \le \alpha + (1-2\alpha)\theta + 2\sqrt{\alpha(1-\alpha)\theta(1-\theta)}.} \tag{12}$$

The proof uses a dimension-to-list conversion. At boundary j, pull the terminal AUDIT measurement backward through the future comb, average over the independent future inputs, and relabel the final syndrome as the cumulative prefix syndrome. This produces a valid POVM on the d_j-dimensional cut memory with the same global success P_{A}. A dimension-d_j POVM implies a classical list L_j of at most d_j labels whose event probability is at least P_{A}.

For a complete transcript, intersect the m causal list events. Their failure probability is at most $m\epsilon$. The cumulative block labels are in bijection with the individual block syndromes, and full rank makes every fibre uniform. The intersection therefore contains at most k computational words. Schur–Horn majorisation turns its coordinate mass into the top-k spectral mass of G_c, proving (11). This route avoids the extra square-root losses of approximate-injectivity arguments.

To obtain RETURN, split the singular values above and below rank k:

$$\operatorname{Tr}\sqrt{G_c} \le \sqrt{k(m_c - t_c)} + \sqrt{(D-k)t_c}, \qquad m_c = \operatorname{Tr} G_c. \tag{13}$$

Instrument completeness gives $\sum_c m_c = D$. Cauchy–Schwarz sums (13), while Ky Fan anti-norm superadditivity gives the universal tail ceiling $\sum_c t_c \le D - k$ [15]. Substitution yields (12). The right-hand side is a squared overlap of two real unit vectors, so its monotonicity on the allowed interval is immediate.

6 An explicit interior order gap

For H_{I}, $m = 2$ and $\alpha = 1/4$. Hence, with

$$\theta = \min\{2(1-P_{\mathrm{A}}), 3/4\}, \tag{14}$$

every strategy satisfies

$$F_{\mathrm{R}} \le \frac{1}{4} + \frac{1}{2}\theta + \frac{\sqrt{3}}{2}\sqrt{\theta(1-\theta)}. \tag{15}$$

Near perfect AUDIT this becomes

$$F_{\mathrm{R}} \le \frac{1}{4} + (1-P_{\mathrm{A}}) + \sqrt{\frac{3}{2}(1-P_{\mathrm{A}})(2P_{\mathrm{A}}-1)}. \tag{16}$$

The square-root exponent is optimal: a legal weak-record family has a $\sqrt{1-P_{\mathrm{A}}}$ correction. The leading constant is not claimed sharp.

Maximizing (15) in the support direction gives a closed certificate.

Corollary 6.1 (Certified interior separation)**.** *For every $0 \le \lambda \le 1$,*

$$\beta^{\mathrm{stream}}_{H_{\mathrm{I}},2}(\lambda) \le U_{\mathrm{I}}(\lambda) = \frac{1}{2} + \frac{\lambda}{4} + \frac{1}{4}\sqrt{7\lambda^2 - 10\lambda + 4}. \tag{17}$$

Moreover,

$$U_{\mathrm{I}}(\lambda) < B_{4,2}(\lambda) \quad \textit{for} \quad \frac{3}{7} < \lambda < 1. \tag{18}$$

Set $\theta = \sin^2\phi$, $0 \le \phi \le \pi/3$. The support becomes a positive 2×2 quadratic form in $(\cos\phi, \sin\phi)$; its largest eigenvalue is (17). Eliminating the two radicals when comparing (17) with (4) gives $-4\lambda^2(\lambda-1)(7\lambda-3)$, with the unsquared signs fixing the interval in (18).

At balanced weight,

$$\beta^{\mathrm{stream}}_{H_{\mathrm{I}},2}(1/2) \le \frac{5}{8} + \frac{\sqrt{3}}{8} = 0.841506350946\ldots, \quad (19)$$

whereas the grouped value is $B_{4,2}(1/2) = 0.853553390593\ldots$. This proves an interior gap without assuming QND, Clifford, projective, covariant, or binary-outcome instruments.

7 Exact leaf reduction and compact construction

Write a fine transcript leaf as $K : A_1 \cdots A_4 \to B_1 \cdots B_4 M$. Its vectorisation is an open-boundary MPS whose three temporal bond ranks are at most two. For a normalised leaf, define

$$A_H(K) = \max_{\{E_s\}} \sum_x \mathrm{Tr}\left[E_{Hx}\,\mathrm{Tr}_{B^4} K|x\rangle\!\langle x|K^\dagger\right], \quad (20)$$

$$R(K) = \frac{1}{16}\left(\mathrm{Tr}\sqrt{K^\dagger K}\right)^2, \qquad \mathrm{Tr}\,K^\dagger K = 1. \quad (21)$$

Theorem 7.1 (Homogeneous Choi-MPS reduction)**.** *For the interleaved game,*

$$\beta^{\mathrm{stream}}_{H_{\mathrm{I}},2}(\lambda) = \max_{\substack{\mathrm{Tr}\,K^\dagger K = 1\\ \text{TT-rank}_i(|K\rangle\!\rangle)\le 2}} \left[\lambda A_{H_{\mathrm{I}}}(K) + (1-\lambda)R(K)\right]. \quad (22)$$

Four classical outcomes per slot suffice.

The upper direction is a homogeneous averaging argument: every complete strategy is a probability distribution over its normalised fine leaves, and both branch scores average with the same leaf weights. For the converse, put an arbitrary feasible leaf in row-isometric MPS gauge, with local tensors T_i satisfying

$$\mathrm{Tr}_{A_i}(T_i^\dagger T_i) = \mathbb{I}_{M_{i-1}}. \quad (23)$$

For each Pauli $P \in \{I, X, Y, Z\}$ use the local Kraus map $T_i(P\otimes\mathbb{I})/\sqrt{2}$. The Pauli one-design identity makes these four maps complete. Every four-slot transcript is K right-multiplied by a tensor product of Paulis and scaled by $1/4$. RETURN is invariant, while the known Pauli bit flips merely relabel the linear syndrome in AUDIT. Thus all 256 leaves have the same homogeneous score, proving (22).

A closed-form construction inside (22) has three nonzero rank-one effects and admits a two-variable support formula. Put

$$\begin{aligned}
g &= 1 - t/2, & z &= -t/(2-t),\\
e &= t(1+r)/2,\\
\Delta &= \sqrt{(1-z^2)(1-r^2)},\\
h &= \frac{g}{2}(1 + zr + \Delta),\\
c_0 &= \sqrt{t} + \sqrt{1-t}, & c_1 &= \sqrt{e} + \sqrt{1-e},
\end{aligned} \quad (24)$$

and let $c_\varnothing = \sqrt{2}$ for $t \ge 1/2$, while $c_\varnothing = c_0$ for $t < 1/2$. Define

$$\begin{aligned}
\boldsymbol{c} &= (c_0, c_\varnothing, \sqrt{2}c_1)^\mathsf{T},\\
M_\lambda(t,r) &= \lambda\,\mathrm{diag}(t, 0, h) + \frac{1-\lambda}{8}\boldsymbol{c}\boldsymbol{c}^\mathsf{T}.
\end{aligned} \quad (25)$$

Then the physical lower bound is

$$L_{\mathrm{I}}(\lambda) = \max\left\{1 - \frac{\lambda}{2}, \max_{\substack{0\le t\le 1\\ -1\le r\le 1}} \lambda_{\max}\big(M_\lambda(t,r)\big)\right\}. \quad (26)$$

The Perron eigenvector fixes the four middle-cut priors. The first two slots prepare those four pure states; slot three implements the three-effect POVM; slot four generates a uniform bit and XORs the binary decision into the terminal qubit. Orthogonal emitted states make $K^\dagger K$ diagonal, so the polar RETURN value in (21) is attained exactly.

At balanced weight, (26) gives

$$\begin{aligned}
t &= 0.458073977\ldots,\\
r &= 0.016373524\ldots,\\
P_{\mathrm{A}} &= 0.620085075586\ldots,\\
F_{\mathrm{R}} &= 0.899520492117\ldots,\\
L_{\mathrm{I}}(1/2) &= 0.759802783851444\ldots.
\end{aligned} \quad (27)$$

It crosses the no-record line at $\lambda_c = 0.441437845098\ldots$. An independent NumPy construction checks all temporal ranks, local row isometries, Pauli completeness, and direct scores; its largest residual is below 9×10^{-16}.

Unrestricted complex Choi-MPS continuation, eight multistarts of the larger pinched cq-instrument relaxation, and a relaxation of the first two slots all agree with (26) on the tested grid. This is substantially stronger evidence than the earlier ternary tree value $0.759448970317\ldots$, and Pauli completion proves that the compact leaf is itself physical. It is still not a global upper certificate for the full nonconvex maximum in (22). The next section closes one support direction, not the whole curve.

8 A certified interior support direction

The homogeneous reduction leaves a finite but nonconvex leaf problem. For a global certificate, combine slots one–two and three–four into two blocks. Let $\rho_z \succeq 0$ be four subnormalised prefix states and let $J_y \succeq 0$ be the input-major Choi matrices of one common four-outcome qubit instrument:

$$\sum_z \mathrm{Tr}\,\rho_z = 1, \qquad \sum_y \mathrm{Tr}_{\mathrm{out}}\,J_y = \mathbb{I}_2. \quad (28)$$

They generate

$$\begin{aligned}
\sigma_{zy} &= \mathrm{Tr}_{\mathrm{in}}[(\rho_z^\mathsf{T}\otimes\mathbb{I}_2)J_y],\\
p_{zy} &= \mathrm{Tr}\,\sigma_{zy},\\
\tau_s &= \sum_{z\oplus y = s}\sigma_{zy}.
\end{aligned} \quad (29)$$

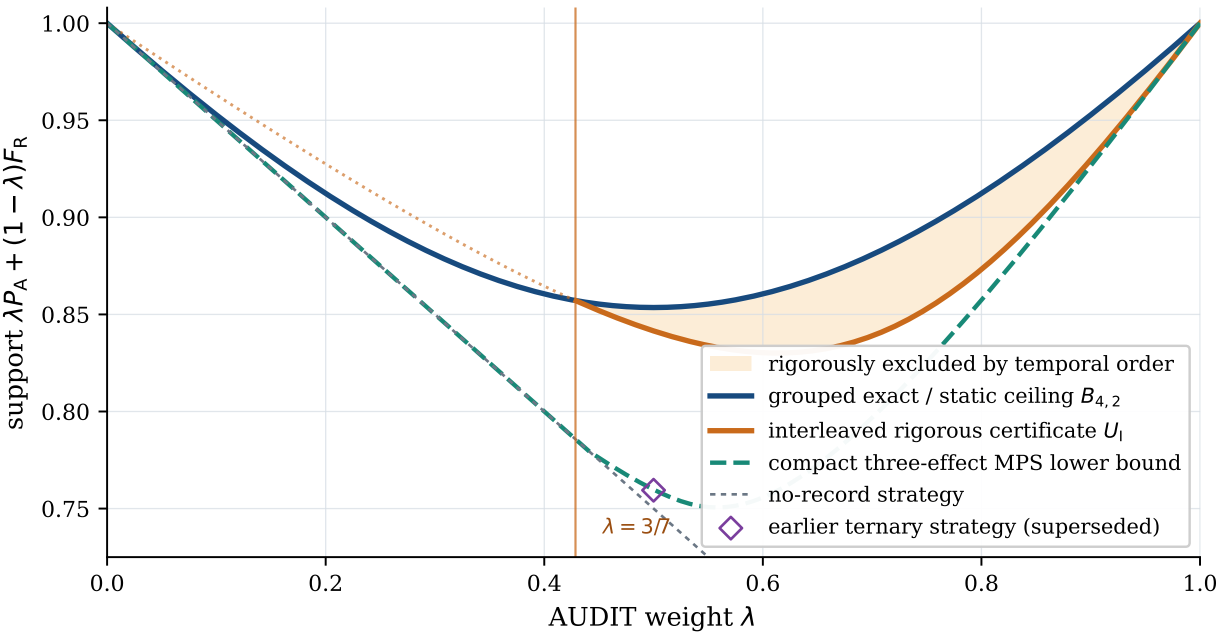

Figure 2: Support values for the same-code pair. The blue curve is the exact grouped frontier. The orange curve is the rigorous interleaved upper certificate; the shaded region is excluded solely by temporal order for $3/7 < \lambda < 1$. The dashed curve is the compact three-effect MPS lower bound; the hollow diamond is the earlier ternary strategy that it supersedes. The unclosed vertical distance is not an error bar: it is the remaining global MPS-maximisation problem.

The common J_y is essential: replacing it by unrelated conditioned states would erase the causal-instrument constraint. Terminal discrimination and computational pinching then give the outer programme

$$\begin{aligned}\beta^{\mathrm{stream}}_{H_1,2}(\lambda) &\le \overline{\beta}_{2\mathrm{b}}(\lambda),\\ \overline{\beta}_{2\mathrm{b}}(\lambda) &= \max_{\rho,J}\Bigg\{\lambda \min_{Y\succeq\tau_s} \operatorname{Tr} Y \\ &\qquad + \frac{1-\lambda}{16}\Bigg(\sum_{z,y}\sqrt{p_{zy}}\Bigg)^2\Bigg\}.\end{aligned} \tag{30}$$

The pinching inequality is the only relaxation in this passage and is tight for the four-effect lower construction.

A linear terminal objective admits an extreme qubit POVM, hence at most four rank-one active effects [16]. The Helstrom conditions make this arity split quantitative. Write $E_i = w_i\Pi_i$, $p_i = \operatorname{Tr}\tau_i$, and $Y = A(\mathbb{I} + t\hat{z}\cdot\sigma)/2$. For an active label,

$$Y - \tau_i = (A - p_i)\Pi_i^{\perp}, \qquad p_i \ge A\frac{1+2tx_i+t^2}{2(1+tx_i)}, \tag{31}$$

where $x_i = \hat{z}\cdot n_i$ and the removable value at $(t, x_i) = (1, -1)$ is zero. POVM completeness gives $\sum_i w_i = 2$ and $\sum_i w_i n_i = 0$. In particular, averaging the binary projective comparison that retains Π_i yields

$$\overline{A}_{2,j} = A - \frac{1-w_j}{2-w_j}(A - p_j). \tag{32}$$

Thus independently certified projective support lines become simultaneous linear restrictions on every fully active readout.

Theorem 8.1 (Certified support at $\lambda = 3/5$)**.** *For the four-slot interleaved game with one persistent qubit,*

$$\boxed{\frac{957373519}{1250000000} \le \beta^{\mathrm{stream}}_{H_1,2}(3/5) \le \overline{\beta}_{2\mathrm{b}}(3/5) \le \frac{7667}{10000}}. \tag{33}$$

Equivalently, the certified decimal interval is $[0.7658988152, 0.76670]$*, of width* 0.0008011848.

Proof. The projective interval kernels certify $\beta_{\mathrm{proj}}(0.55) \le 0.7573$ and $\beta_{\mathrm{proj}}(0.60) \le 0.7660$. Equations (31)–(32), the polygon closure of the transverse Bloch components, and McCormick envelopes on exact rational weight boxes reduce the remaining cases to the five rows of Table 1. Their replayed maxima are respectively below the displayed outward endpoints. If a four-outcome effect has trace below $\delta = 0.0003$, merging that outcome into a retained answer leaves RETURN unchanged and reduces AUDIT by at most δ. The last row is therefore bounded by $0.76652 + (3/5)\delta = 0.76670$, which is the largest sector bound.

For the lower endpoint, five rational half-angle coordinates specify the symmetric four-effect Choi MPS, including its priors. They normalise every state and effect identically, while the four effects sum exactly to $\mathbb{I}_2$. AUDIT is rational. Flooring the four square roots in polar RETURN to 192 dyadic bits gives a rational support lower bound $0.76589881526469403 19\ldots >$ 0.7658988152. Local Pauli completion makes the leaf a complete causal instrument, proving the first inequality. □

The projective proof kernel uses directed IEEE-754 operations expanded with `nextafter`; its deterministic

Table 1: Exhaustive terminal-readout certificate at $\lambda = 3/5$. A conic "exact replay" reconstructs the stated cone, repairs dual cone membership, and evaluates the residual correction and objective with rational arithmetic. CLARABEL proposes dual vectors during generation but is not called or trusted during replay.

Sector	proof kernel	units replayed	certified upper
at most two active effects	outward interval cover of four projective topologies	1,448 cells	0.76600
three or four active, $w_{\max} \leq 0.88325$	capped-weight SOCP exact dual replay	576 cells	0.765893818
three active, $w_{\max} > 0.88325$	transferred ternary SOCP exact dual replay	12,008 cells	0.76652
four active, $w_{\max} > 0.88325$, $w_{\min} \geq 0.0003$	McCormick–SOCP exact dual replay	90×6 orders	0.76670
four active, $w_{\min} < 0.0003$	delete the smallest effect into the ternary sector	exact inequality	$0.76652 + 0.00018$

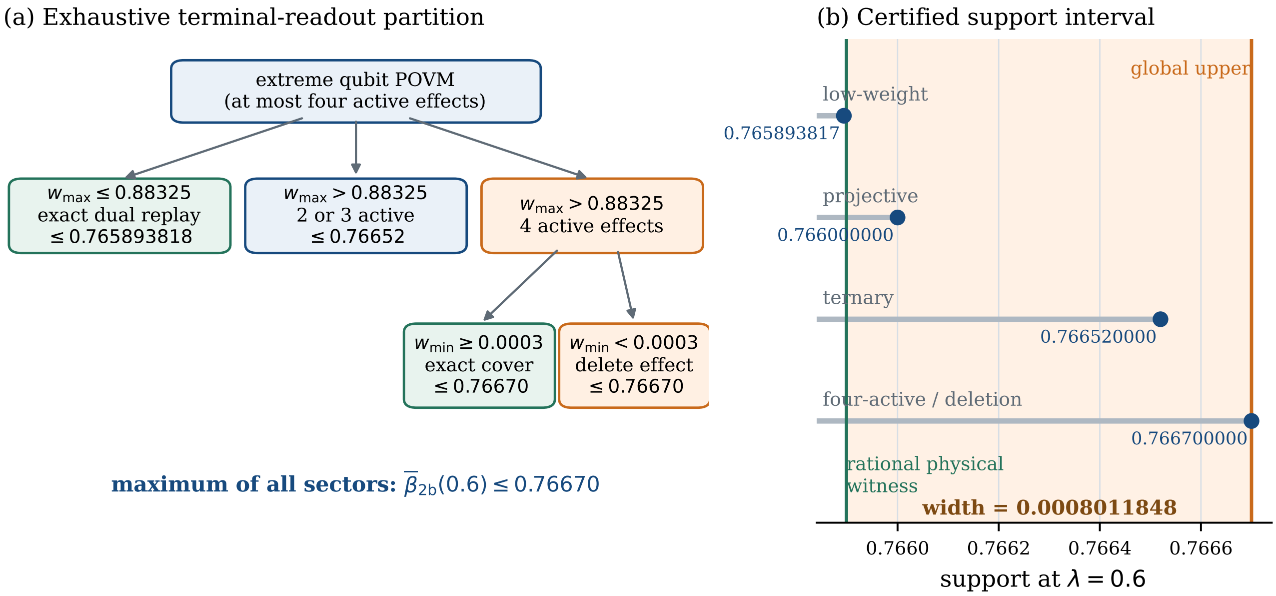

Figure 3: How the global direction is closed. Panel (a) shows the exhaustive arity and weight partition; the deletion branch is the dominant outward endpoint. Panel (b) separates the independently certified sector caps from the rational physical witness. The shaded strip is a theorem interval, not a statistical error bar.

generators must be rerun to reproduce the compact cell summaries. The three conic replays use exact fractions and call no optimiser, but still trust the Python reconstruction of the cone matrices. Lean checks the analytic support-cut bridges, sector partition, and final rational arithmetic; it does not replay the interval trees or the CVXPY canonicalisation. Thus (33) is solver independent in verification, not a claim of end-to-end kernel formalisation.

9 Relation to neighbouring disciplines

The result sits at a specific intersection. Bounded-storage cryptography and postmeasurement information already turn a quantum dimension into a late decoding limitation [6, 7]. Quantum-comb memory cost already separates persistent coherent dimensions from classical assistance [3, 4]. Sequential state generation identifies persistent ancilla dimension with a temporal tensor-network bond [17]. Trellises, sectionalization, and matroid pathwidth already quantify order-sensitive live constraints in linear codes [11, 14, 13]. Nondestructive discrimination connects classical labels with recoverable entanglement in a spatial LOCC setting [18, 19].

The present bridge is operational rather than terminological. A causal dimension bound is converted into one classical list at every full-rank temporal section. The intersection size controls a Ky Fan tail of each instrument leaf, which in turn controls all-carrier entanglement recovery. None of the nearby papers located in the focused review states this chain for the same-code, late-choice syndrome/return interface. Conversely, the work does not claim that order dependence, trellis width, bounded quantum memory, or information–disturbance is new in isolation.

10 Limits and open exact frontier

The theorem is conditional on a trusted resource boundary. Every coherent system crossing a block cut must count toward d_j; the classical transcript must be dephased; past carriers must be unavailable in AUDIT; and RETURN must include all failures. Relaxing any of these assumptions changes the task. Visible memory reset does not prove that an inaccessible environment contains no record.

The interval (18) resolves the existence of an interior order effect, not the full interleaved Pareto boundary. At balanced weight the best verified lower strategy and rigorous upper certificate are

$$0.759802783851\ldots \leq \beta^{\text{stream}}_{H_1,2}(1/2) \\ \leq 0.841506350946\ldots. \tag{34}$$

The Choi-MPS reduction proves that local completeness and unbounded adaptive outcome trees are no longer the obstruction. At balanced weight, closing the large remaining gap still requires either a better bond-two MPS leaf or a sharper global certificate. At $\lambda = 3/5$, Eq. (33) instead leaves a relative width of approximately 0.10461% but does not identify the exact maximiser. The complete support curve, the optimal coefficient in the near-endpoint square-root law, and extensions to partially ranked block sequences remain open.

Nothing in the observed statistics selects an interpretation of quantum mechanics. The result is a standard operational statement about coherent memory, temporal order, and recoverability. It neither observes alternative worlds nor communicates with an environmentally decohered branch.

11 Conclusion

Permuting the columns of one linear syndrome can change the attainable information–recovery region of a fixed coherent memory. Grouping equal columns allows a one-qubit device to attain the static frontier; interleaving them creates an additional full-rank temporal boundary, squares the perfect-AUDIT recovery penalty, and produces a rigorously certified interior gap. Causal list decoding provides the robust mechanism: approximate terminal knowledge yields small lists at every memory cut, their intersection bounds a spectral tail, and that tail limits coherent recovery. This narrow theorem, rather than a many-worlds narrative, is the contribution that survives the originality audit. Local Pauli completion further removes the arbitrary adaptive tree exactly and leaves one finite bond-two Choi-MPS maximum. The three-effect construction supplies the strongest known balanced lower point; a distinct rational four-effect phase and an independently replayable sector certificate close the direction $\lambda = 3/5$ to width 0.0008011848. Exact equality there and the rest of the support curve are deliberately left open.

A Dimension-to-list lemma

Lemma A.1. *Let $\{\rho_x\}_{x\in\mathcal{X}}$ be subnormalised states on a d-dimensional system and $\{E_x\}_{x\in\mathcal{X}}$ a POVM. If $p_x = \operatorname{Tr}\rho_x$, then*

$$\sum_x \operatorname{Tr}(E_x\rho_x) \leq \max_{L\subseteq\mathcal{X},\,|L|\leq d} \sum_{x\in L} p_x. \tag{35}$$

Proof. Positivity gives $\operatorname{Tr}(E_x\rho_x) \leq p_x\|E_x\|_\infty$. The weights $w_x = \|E_x\|_\infty$ satisfy $0 \leq w_x \leq 1$ and

$$\sum_x w_x \leq \sum_x \operatorname{Tr} E_x = d. \tag{36}$$

The resulting fractional d-unit linear program assigns unit weight to the d largest p_x, proving (35). □

B Pulled-back boundary measurements

Write a word as blocks $x = (x_1, \ldots, x_m)$ and define

$$v_j = H^{(j)}x_j, \qquad u_j = v_1 + \cdots + v_j. \tag{37}$$

Let a_j be the refined transcript through boundary j. Conditioned on a_j and $u_j = u$, trace all sequestered prefix carriers and within-fibre variables, producing a subnormalised state $\rho^{(j)}_{a_j,u}$ on the cut memory.

Fix the future block inputs and compose the suffix comb with the terminal AUDIT POVM. Relabel its final answer u_m as

$$u_j = u_m - (v_{j+1} + \cdots + v_m). \tag{38}$$

Averaging these effects over independent uniform future inputs gives a POVM on the cut memory. Summed over prefix transcripts, its success is the original P_{A}. The construction remains valid when the cut memory is entangled with past carriers: the future comb acts only on the memory and the carriers are traced in AUDIT.

Applying Appendix A at every prefix transcript gives a classical list $L_j(a_j) \subseteq \mathbb{F}_q^r$, $|L_j(a_j)| \leq d_j$, such that

$$\Pr[U_j \in L_j(A_j)] \geq P_{\mathrm{A}}. \tag{39}$$

The POVMs for different j need not be jointly measurable; the lists are proof objects, not simultaneous physical measurements.

For a complete transcript c, define

$$S_c = \{x : u_j(x) \in L_j(a_j(c)) \text{ for all } j\}. \tag{40}$$

The map $(u_1, \ldots, u_m) \leftrightarrow (v_1, \ldots, v_m)$ is bijective. Every full-rank block has fibre size q^{n_j-r}, hence

$$|S_c| \leq q^{n-mr}\prod_j d_j = k. \tag{41}$$

The union bound applied to (39) yields

$$\sum_c \sum_{x\notin S_c} \|K_c|x\rangle\|^2 \leq mD(1-P_{\mathrm{A}}). \tag{42}$$

The diagonal of G_c is precisely $\|K_c|x\rangle\|^2$. Schur–Horn majorisation implies that the diagonal mass on any set of at most k coordinates is no greater than the sum of the top k eigenvalues. Subtracting from $\operatorname{Tr} G_c$ in (42) proves (11).

C From spectral tail to recovery

For a fine-grained leaf, optimal polar correction contributes at most

$$F_{\mathrm{R},c} \le \frac{(\operatorname{Tr}\sqrt{G_c})^2}{D^2}. \tag{43}$$

This already allows a joint decoder on every emitted carrier and terminal memory; dropping the reset projector is an upper relaxation. Applying Cauchy–Schwarz separately to the first k and last $D-k$ singular values gives (13). Let $T=\sum_c t_c=\theta' D$. Since $\sum_c G_c=\mathbb{I}_D$, one has $\sum_c m_c = D$. Summing the square of (13) and applying Cauchy–Schwarz to $\sum_c \sqrt{t_c(m_c-t_c)}$ gives

$$F_{\mathrm{R}} \le \alpha + (1-2\alpha)\theta' + 2\sqrt{\alpha(1-\alpha)\theta'(1-\theta')}. \tag{44}$$

The sum of the smallest $D-k$ eigenvalues is a homogeneous concave spectral functional, hence superadditive. Therefore

$$T \le \sum_{\ell>k} \lambda_\ell\left(\sum_c G_c\right) = D-k. \tag{45}$$

Together with (11), this gives $0 \le \theta' \le \min\{m(1-P_{\mathrm{A}}), 1-\alpha\}$. The recovery expression is increasing throughout that interval, proving (12).

D Homogeneous Choi-MPS completion

This appendix proves (22). Refine every classical transcript of a physical four-slot strategy until it has one global Kraus operator K_c, and put

$$G_c = K_c^\dagger K_c, \qquad t_c = \operatorname{Tr} G_c. \tag{46}$$

Instrument completeness gives $\sum_c G_c = \mathbb{I}_{16}$ and $\sum_c t_c = 16$. For $t_c>0$, write $\widehat{K}_c = K_c/\sqrt{t_c}$. Because the terminal measurement and flagged recovery may depend on c, the two branch scores decompose exactly as

$$P_{\mathrm{A}} = \sum_c \frac{t_c}{16} A_{H_{\mathrm{I}}}(\widehat{K}_c), \qquad F_{\mathrm{R}} = \sum_c \frac{t_c}{16} R(\widehat{K}_c). \tag{47}$$

The weights in (47) sum to one. Every leaf has temporal TT ranks at most two, so its support score is no larger than the right-hand side of (22). This proves the upper direction independently of the number or adaptivity of the outcomes.

For the converse, take any normalised operator K admitted by the maximum in (22). Successive Schmidt decompositions put its vectorisation in row-isometric MPS gauge. Its local tensors

$$T_i : A_i \otimes M_{i-1} \longrightarrow B_i \otimes M_i \tag{48}$$

obey

$$\operatorname{Tr}_{A_i}(T_i^\dagger T_i) = \mathbb{I}_{M_{i-1}}. \tag{49}$$

Zero Schmidt subspaces may be removed and smaller bonds padded, so this gauge does not increase the allowed memory dimension.

For $P\in\{I,X,Y,Z\}$, define the four local Kraus maps

$$L_{i,P} = 2^{-1/2} T_i (P\otimes \mathbb{I}_{M_{i-1}}). \tag{50}$$

The Pauli one-design identity

$$\sum_P (P^\dagger \otimes \mathbb{I}) Y (P\otimes \mathbb{I}) = 2\mathbb{I}_{A_i} \otimes \operatorname{Tr}_{A_i} Y \tag{51}$$

together with (49) yields $\sum_P L_{i,P}^\dagger L_{i,P} = \mathbb{I}_{A_i M_{i-1}}$. Thus (50) is a complete causal instrument at every slot and its outcome may be dephased immediately.

For a full Pauli transcript $\boldsymbol{P}=(P_1,\ldots,P_4)$, contraction gives

$$K_{\boldsymbol{P}} = \frac{1}{4} K(P_1\otimes\cdots\otimes P_4). \tag{52}$$

All 256 leaves have $\operatorname{Tr} K_{\boldsymbol{P}}^\dagger K_{\boldsymbol{P}} = 1/16$ and hence probability 1/256 on the maximally mixed input. Right-unitary invariance preserves $R(K)$. If $P_i = X^{a_i} Z^{b_i}$, then $P_i|x_i\rangle$ is a phase times $|x_i+a_i\rangle$; subtracting the known shift $H_{\mathrm{I}}(a_1,\ldots,a_4)$ from the terminal answer preserves $A_{H_{\mathrm{I}}}(K)$. Every conditional leaf therefore has the same two scores as K. Their complete mixture attains its homogeneous quotient and proves the lower direction of (22).

E Reproducibility and claim boundary

The code, stored instrument, tests, literature ledger, and both manuscript sources are public at `https://github.com/asim-v/carmenq`. The commands

```
python -m pytest -q
python scripts/verify_interleaved_counterexample.py
python scripts/verify_compact_interleaved_candidate.
py
python scripts/verify_four_effect_rational_lower.py
python scratch/d2_frontier/verify_global_frontier_
1060.py
python scripts/generate_order_gap_figure.py
python scripts/generate_global_frontier_figure.py
python scripts/build_order_pdf.py
```

reproduce the software checks, physical constructions, certificate assembly, plotted bounds, and this PDF. Numerical searches are explicitly lower-bound generators. The proved claims are (8), (11), (12), (22), the attainability of (26), and the intervals (18) and (33). Equality between (26) and the exact interleaved interior curve is not claimed.

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